Few-Shot Class-Incremental SAR Target Recognition via Cosine Prototype Learning

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***Abstract*— Recent years have witnessed a remarkable breakthrough in synthetic aperture radar automatic target recognition (SAR ATR) with the development of deep learning (DL). Nonetheless, once deployed, the DL-based methods' ability to incrementally learn new knowledge from few-shot samples without forgetting the old is fragile, hindering them from discriminating unseen targets in real-world situations. In this article, we propose a cosine prototype learning (CPL) framework to first unlock few-shot class-incremental learning (FSCIL) in the SAR ATR field inspired by the intrinsic relationships between target azimuth-aware knowledge and semantic features under the cosine criterion. By condensing class-specific characteristics into individual prototypes, stable profiles of targets are depicted without losing generalization. For the model's plasticity, a pairwise structure separation (PSS) loss is introduced to separate old and new classes and compact intraclass features. Meanwhile, the model's transferability on new classes is guaranteed by a prototype consistency (PC) loss. For the model's stability, we propose a prototype–exemplar distillation (PED) loss and a prototype recalibration (PR) strategy to penalize semantic drifts of old-class feature spaces and alleviate the misalignment of the learned prototypes successively. At inference, a nearest-class-mean (NCM) classifier is adopted for evaluation by comparing cosine similarity scores between testing samples and class-specific prototypes. In experiments, the proposed components of our method are explored by ablation studies. Strong baselines are established, and extensive experiments conducted on the moving and stationary target acquisition and recognition (MSTAR) dataset show that our method outperforms state-of-the-art methods under various FSCIL conditions, verifying its effectiveness for the FSCIL of SAR ATR.**

***Index Terms*— Catastrophic forgetting, cosine prototype learning (CPL), cosine structures, few-shot class-incremental learning (FSCIL), SAR automatic target recognition (ATR), synthetic aperture radar (SAR).**

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I. Introduction

As a longstanding, fundamental, and challenging problem in synthetic aperture radar (SAR) imagery interpretation, SAR automatic target recognition (SAR ATR) has been a topic of intensive research for several decades due to its indispensable role in a wide variety of civil and military applications, including military surveillance [1], [2], [3], [4], [5] and border security [6], [7]. Recently with the rapid advancement in SAR imaging technique, the amount of high-resolution SAR images is growing exponentially, enabling increasingly more research opportunities for challenging SAR target recognition tasks. Numerous efforts, including traditional and recent deep learning (DL) methods, have been devoted to the problem of SAR ATR, and considerable advances have been achieved. However, there is still a long way to go in improving the performance of SAR ATR systems to satisfy real-world scenarios.

Needless to say, DL, with powerful data representing capability, has brought tremendous progress for the problem of object recognition in computer vision and SAR ATR fields, e.g., incredible accuracy for recognizing a thousand different classes of objects in natural images [8] or classifying targets and their variants imaged at various radar configurations [9], [10]. It also brings both opportunities and challenges. Therefore, in the past few years, the research community has devoted itself to moving toward many more challenging settings of object recognition, such as low-shot learning [11], [12], [13] and class-incremental learning (IL) [14], [15]. Recently, few-shot class-IL (FSCIL) [16], [17], [18], [19] has attracted significant attention. Shortly, FSCIL aims to address the problem of incrementally learning new knowledge from a few labeled examples without forgetting the old [16]. It is a realistic yet challenging issue in many real-world scenarios and is particularly demanded to be solved in the SAR ATR field due to the apparent contradictions between characteristics of the DL-based methods and real-world situations.

1) *Dynamic World and Frozen Models:* The real-world SAR ATR situations are dynamic in which target information, such as sample and class, is observed gradually over a long period. However, once deployed, the current DL-based methods could recognize target classes learned at the initial training stage but hardly update from newly collected concepts in time due to their fixed parameters. Moreover, the DL-based models must be retrained from scratch by using all training data to discriminate both old and new classes, which is

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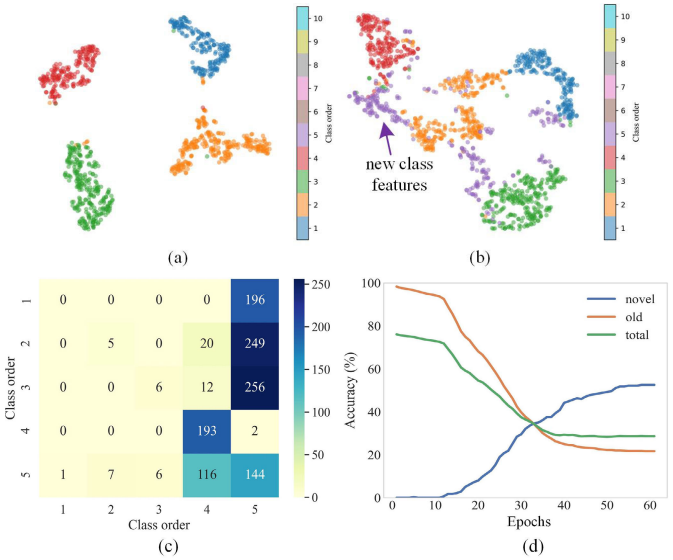


Fig. 1. DL-based SAR ATR method encounters serious catastrophic forgetting of old knowledge when learning a new class. (a) t-SNE of features of the four base classes. (b) t-SNE of features after adaption to a new class. (c) Confusion matrix of total classes. (d) Accuracy curves of total, old, and new classes during learning.

privacy-destroyed, time-consuming, and even unreachable at resource-restricted SAR ATR platforms, such as mobile or edge devices.

2) *Limited Knowledge and Data-Driven Models:* The real-world SAR ATR situations are usually noncooperative, in which acquiring abundant and diversified target samples is tough and impractical. Meanwhile, annotating objects in SAR images by human beings is highly expensive, as only SAR experts can do the labeling due to the particular characteristics of SAR imagery. However, the DL-based methods are data-oriented, and their competitive performance heavily relies on the support of enormously high-quantity labeled data.

Thus, it is essential to equip SAR ATR systems with the capability of incrementally learning knowledge of new classes from limited labeled examples without losing performance on the old. Current works of FSCIL mostly focus on evaluating natural images. There has been no attempt to investigate FSCIL in the field of SAR ATR, and its potential remains locked. Hence, unlocking the FSCIL of SAR ATR and exploring its unique and practical issues are urgently necessary.

As aforementioned, the DL has been broadly used in the SAR ATR field, benefiting from its powerful representing ability. However, directly applying the DL-based methods for FSCIL inevitably leads to the notorious **catastrophic forgetting** problem, seriously exacerbating the performance of SAR ATR. As a toy example shown in Fig. 1, feature spaces of a DL-based method trained on four classes encounter a serious distortion when learning a new class. As a result, the confusion matrix is biased toward the new class, and a serious performance drop (PD) on both old and total classes appears if no special techniques are adopted. Striking the problem resorts to addressing the **stability–plasticity dilemma**: stability refers to the preservation of learned knowledge, and plasticity refers to the integration of the new while learning it continually [14].

Furthermore, the FSCIL of SAR ATR is much more intractable due to prominently domain-specific challenges demonstrated as follows.

1) *Extremely Poor Information of Targets:* Targets in SAR images are roughly represented as backscattering points with specific topological patterns resulting from the particular imaging mechanisms of SAR. Still, they lack abundant cues of color and appearance, providing limited guidance for network optimization. Therefore, capturing representative features of targets in SAR images calls for in-depth explorations of structural information of their backscattering points.
2) *Seriously Interclass Similarity and Intraclass Variation:* Due to the diversity of imaging conditions and the similarity of target configurations, such as target azimuths and components, e.g., turrets equipped by T72 and T62, features of different classes are easily confused, and discrepancies within the same classes might be pronounced in some situations. Thus, maintaining the model's discrimination on targets at the IL process calls for paying more attention to targets lying at decision boundaries and enhancing the feature consistency of intraclass samples.
3) *Uninformative Background Interference:* Due to the diversity and nonhomogeneity of surroundings as well as the particularity of SAR imaging mechanisms, intensive clutters might widely exist around targets and cause much interference to the DL-based models for representing and discriminating target characteristics accurately, especially for FSCIL of SAR ATR. Thus, identifying targets in SAR images calls for designing a target-oriented and noise-insensitive classifying criterion.

In response to the aforesaid dilemma and domain-specific challenges, a cosine prototype learning (CPL) framework is proposed to solve the FSCIL of SAR ATR. Inspired by the intrinsic relationships between target azimuth-aware knowledge and semantic features under the cosine criterion, class-aware knowledge is represented as cosine prototypes to fully capture targets' general and stable profiles. The model's capability of learning new concepts without forgetting the old is guaranteed by various structural losses at the cosine space.

Most specifically, at the base learning stage, where a large amount of training data is provided in advance, a convolutional neural network (CNN)-based backbone is trained by a cosine cross-entropy (CE) loss for extracting target general features at later incremental phases. The model's plasticity for learning structural cues of new targets without losing generalization is guaranteed by a pairwise structure separation (PSS) loss and a prototype consistency (PC) loss. To preserve old knowledge, a prototype–exemplar distillation (PED) loss and a prototype recalibration (PR) strategy are introduced by penalizing semantic drifts of old feature spaces and rectifying discrimination of the corresponding prototypes successively. At inference, a nearest-class-mean (NCM) classifier is adopted to assign testing samples with classes sharing the largest cosine similarity among saved prototypes of all classes.

In experiments, the contributions of components in the CPL are verified by ablation studies. Our method is compared with

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several baselines at various FSCIL settings on the moving and stationary target acquisition and recognition (MSTAR) dataset. Extensive results show that the proposed method, exploring relationships between target structural cues and cosine semantic features, significantly outperforms all the compared methods by a large margin, verifying its effectiveness of FSCIL in practical SAR ATR scenarios.

To summarize, our main contributions are threefold.

1) We first unlock the FSCIL of SAR ATR and propose a CPL framework by exploring intrinsic relationships between target structural cues and cosine semantic features.
2) We verify the advantages of the cosine criterion for representing target topological cues and address the stability and plasticity dilemma and domain-related challenges by designing three structural losses, i.e., PSS, PC, and PED, and a PR strategy under the criterion.
3) Our method achieves competitive performance and outperforms various state-of-the-art methods by a large margin at the MSTAR dataset under various FSCIL settings.

The rest of this article is organized as follows. We discuss some related work about SAR target classification, IL, and FSCIL in Section II. In Section III, the problem setting of FSCIL, motivations, and the proposed method are introduced in detail. Experiments and analysis are provided in Section IV. Section V concludes the work of this article.

II. Related Works

A. SAR Target Classification

As a high-level task of SAR ATR, numerous approaches have been proposed for SAR target classification in the past several decades, which could be broadly taxonomized into template-based [20], [21], [22], model-based [23], [24], [25], [26], and machine learning-based [27], [28], [29] paradigms. Concisely, template-based methods generally identify targets by computing correlations between testing samples and pre-accumulated sets of class templates, which are easily influenced by target configurations and background clutters. Benefiting from the particular consideration of prior knowledge, such as target electromagnetic (EM) characteristics and background statistical distributions, model-based methods are stable. However, these methods always involve an unacceptable computing burden. Recently, benefiting from the powerful feature representation capability, machine learning methods, especially the DL-based methods [9], [30], [31], [32], [33], have dominated the field and suppressed almost all the previous solutions. However, their remarkable performance greatly depends on a large amount of labeled data and a tedious training process, which are unrealistic in most SAR ATR applications where only few-shot samples can be acquired continually.

B. Incremental Learning

IL, also known as continual learning (CL) or lifelong learning (LL), aims to train a model to learn novel knowledge continually without forgetting the old. An inherent issue of IL is *catastrophic forgetting*, which means that the performance of a pretrained DL-based method may encounter a sharp decrease in classifying old classes when learning the new. The existing work for overcoming the problem can be roughly divided into regularization-based and distillation-based methods [14]. Among the two solutions, competitive performance can be achieved easily by the latter [34], [35], in which old knowledge is preserved by keeping the consistency of outputs, such as logits and features, of old classes generated by the former fixed and the current trainable models [36], [37], [38]. However, the effective preservation of old knowledge in these methods largely depends on accumulating large-scale training samples at the current stage, which is impractical in practical SAR applications. Therefore, it is still essential to equip the DL-based models with an ability to learn from limited samples continually without forgetting the old in real-world SAR ATR tasks.

C. Few-Shot Class-IL

FSCIL is a challenging but practical problem, which requires CNN models to incrementally learn new classes from very few labeled samples without forgetting the previously learned ones [16]. Due to insufficient supervision provided by the current task, problems of catastrophic forgetting of old knowledge and inaccurate representation of the new are prominent. Previous works solving the issue could be categorized into distillation-based [17], [18], [39], weight-regularization-based [40], [41], [42], [43], [44], and meta-learning-based [19], [45], [46] schemes. Derived from IL, distillation-based methods usually store a few representative exemplars of former classes and utilize them to provide supervision signals at various aspects, e.g., logits and features, to the current model. Weight-regularization-based methods defy forgetting by restricting drifts of important model weights between the frozen old and the current trainable models. In meta-learning-based methods, models are trained to learn from few-shot samples continually after numerous episodic tasks at the meta-training stage. At a meta-testing stage, unseen samples could be recognized by constructing similar few-shot incremental episodes as those in the training stage.

Our method belongs to the distillation-based paradigm and is the first to unlock the FSCIL of SAR ATR by deeply exploring domain-specific knowledge at cosine feature space.

III. Methodology

In this section, the problem setup of FSCIL is first provided followed by a detailed description of our motivations. Then, the overall framework is provided with in-depth introductions of the proposed losses and strategy.

A. Problem Setup

FSCIL: It aims to learn new tasks incrementally from a small number of labeled samples without forgetting the previously learned ones [16]. Assume there is a sequence of labeled data $\{D^1, D^2, \ldots D^t\}$, where $D^t = \{(x_i^t, y_i^t)\}_{i=1}^{|D^t|}$ contains $|D^t|$ training instances at session t. x_i^t is an example

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of class $y_i^t \in \mathcal{C}^t$. $\mathcal{C}^t$ is class set of session t. Notably, there is no overlap in class sets among different sessions, i.e., $\mathcal{C}^j \cap \mathcal{C}^k = \varnothing$, for $\forall\ j,k \in t, j \neq k$. Usually, the base session D^1 contains numerous classes, and a large amount of training data is provided for each class. The incremental sessions D^t $(t > 1)$ contain a sequence of classes with few-shot examples and are often provided as an N-way K-shot form with small N and K in each session. At session t, only new data D^t and probably a few exemplars of learned classes until session $t-1$, notated as E^{t-1}, are accessible for optimization. After learning, the trained CNN model is evaluated over all seen categories $\mathcal{C} = \cup_{i=1}^{t} \mathcal{C}^i$ until session t.

B. Motivations

In this section, the advantages of representing target structural cues under the cosine criterion are explored, followed by a description of the relationship between the criterion and the FSCIL of SAR ATR.

1) *Advantages of Cosine Criterion:* Traditionally, the posterior probability of a sample x can be calculated as (1), where $\phi(x)$ is the semantic features of x mapped by a CNN ϕ. $\mathbf{v}_c$ is the weight of class c. C is the total number of classes

$$p(c \mid x) = \frac{\exp(\kappa(\phi(x), \mathbf{v}_c))}{\sum_{j=1}^{C} \exp\left(\kappa\left(\phi(x), \mathbf{v}_j\right)\right)}. \tag{1}$$

In comparison with judging targets by the fully connection (FC) and Euclidean criteria, where $\kappa(\phi(x), \mathbf{v}_c) = \mathbf{v}_c^{\intercal}\phi(x)$ and $\kappa(\phi(x), \mathbf{v}_c) = -\|\phi(x) - \mathbf{v}_c\|^2$, respectively, identifying targets by cosine can be formulated as (2), of which the inherent advantages are threefold

$$\kappa(\phi(x), \mathbf{v}_c) = \frac{\mathbf{v}_c^{\intercal}\phi(x)}{\|\phi(x)\|\|\mathbf{v}_c\|}. \tag{2}$$

First, target azimuth-aware topological structures can be precisely captured and represented by cosine semantic features. For example, with a boyevaya mashina pekhoty (BMP2) vehicle imaged at a 45° azimuth in the MSTAR dataset as an anchor, the structural similarity index measure (SSIM) of foregrounds, i.e., target areas, between the anchor and the same class samples with azimuths ranging from 0 to 360 is given by Fig. 2(a). The curve of SSIM fluctuates with a certain periodicity as azimuths change. Correspondingly, activation values, such as similarity scores, logits, and distances under cosine, FC, and Euclidean criteria, of semantic features between the anchor and other samples are shown in Fig. 2(b)–(d), separately. Because of the special attention on target phase signals by cosine criterion, target azimuth-aware structures are fully captured, exhibiting a tight consistency of similarity scores between the SSIM and semantic features. However, due to the influence of amplitude outliers and insufficient representation of target mutual scattering patterns, ambiguous relationships exist between the SSIM and target semantic features under FC and Euclidean criteria.

Second, target salient components can be represented accurately by cosine semantic features. In Fig. 3, components of two targets at different azimuths are represented

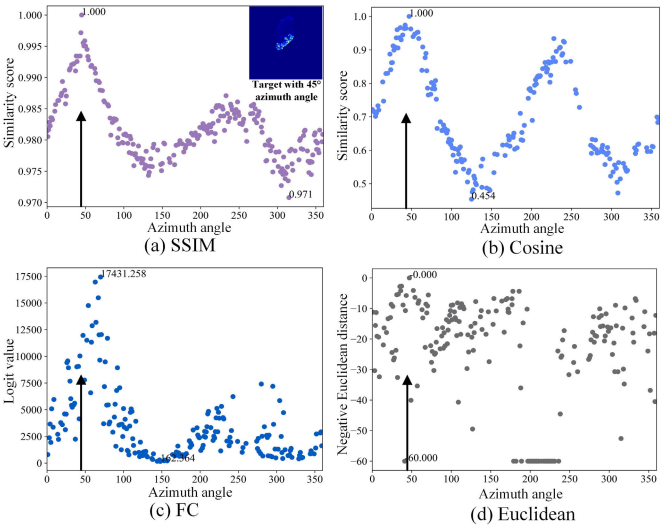


Fig. 2. Correlations of pixel-level structures and semantic activations between an anchor target at 45° azimuth and class targets covering full azimuths. (a) Similarity of pixel-level SSIM. (b) Similarity among features under cosine criterion. (c) Logit activations among features under FC criterion. (d) Negative Euclidean distances among features under Euclidean criterion.

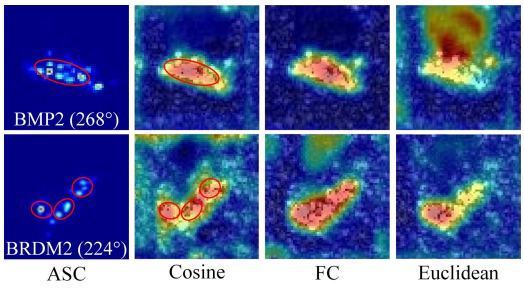


Fig. 3. Relationships between target salient ASCs and Grad-CAM of their semantic features under the three criteria. The strong backscattering parts of BMP2 and BRDM2 ground vehicles in SAR images distribute densely and sparsely, respectively.

by attribute scattering center (ASC) model [47] (the first column of Fig. 3). Meanwhile, gradient-weighted class activation Mapping (Grad-CAM) [48] results of salient features at the three feature spaces are also shown in the last three columns of Fig. 3. It could be seen that locations and amplitudes of target salient features under the cosine criterion (see the red ellipses for examples) are highly consistent with those of their ASCs at various azimuths (see scattering points of BMP2 and boyevaya razvedyvatelnaya dozornaya mashina (BRDM2) for examples).

Third, target characteristics represented under the cosine criterion are robust to background interference. The performance of a DL-based method for classifying ten-class vehicles under the standard operation conditions (SOCs) of the MSTAR dataset is shown in Fig. 4(a). As the peak signal-to-noise ratio (PSNR) of testing images decreases, the performance obtained under the cosine criterion (the blue bar) is consistently better than that under FC and Euclidean criteria. Besides, the impact of surroundings on a CNN-based classifier trained by pure targets is explored. In Fig. 4(b), the classifier trained by cosine criterion is superior to classifiers trained by FC and

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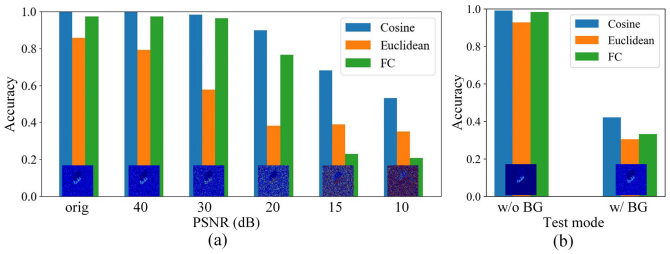


Fig. 4. Robustness of the three criteria to background noise. (a) Classification performance of DL-based method with the degradation of image quality. (b) Robustness of the DL-based method trained by foreground targets and tested by samples with backgrounds.

Euclidean criteria for identifying targets without or with background interference, i.e., w/o BG and w/ BG, verifying the robustness of cosine criterion to background interference.

2) *Relationship Between Cosine Criterion and FSCIL of SAR ATR:* For FSCIL of SAR ATR, the model's ability to learn new concepts from few-shot samples without forgetting the previous ones calls for a trade-off of the model's plasticity and stability. In particular, considering the scarcity of discerning target information provided by few-shot instances and the serious background interference, detailed information of target azimuth-aware backscattering structures should be fully captured to strike the forgetting issue and learn new concepts adaptively. Because of the above advantages of cosine criterion, it is reasonable to construct a cosine-based framework for FSCIL of SAR ATR to dynamically preserve class-aware discerning structures and discriminate targets of different classes, especially those imaged under similar conditions, e.g., components and azimuths.

C. Overall Framework

The overall framework of the proposed method is shown in Fig. 5, which includes base training, incremental training, and session testing stages.

During the base training stage, a CNN feature extractor parameterized as ϕ is first trained on the base session data D^1 by a cosine CE loss for discriminating base session classes and extracting general characteristics of new classes at later incremental sessions. Then, the trained feature extractor ϕ^b, class prototypes $p_1, p_2, \ldots, p_{c^1}$ calculated by averaging features of class-aware training samples of D^1, and the saved exemplars E^1 of the class set $\mathcal{C}^1$ selected by the herding strategy [36] are saved for later learning. At the incremental training stages, for example, the first incremental session ($t = 2$), the few-shot novel dataset D^2, and the saved exemplars E^1 are combined as training data of the current session, which includes classes ranging from 1 to c^2. During IL, PSS and PC losses are utilized to enlarge discrepancies between the novel and other classes and alleviate the model's overfitting to specific characteristics of novel classes, respectively. The PED loss is introduced to preserve the knowledge of old classes, especially the structural relationships, by regularizing distortions of old-class feature spaces mapped by the former fixed and the current trainable models. After learning, prototypes of both new classes of $\mathcal{C}^2$ and old classes $\mathcal{C}^1$ are recalibrated by a PR strategy. A few representative exemplars of $\mathcal{C}^2$ are selected by the herding strategy for data replay at later sessions. By repeating the above steps until the last session t, a trained CNN-based feature extractor ϕ^t and prototypes of classes $\mathcal{C} = \cup_{i=1}^{t}\mathcal{C}^i$ are obtained finally. At the testing stage, for example, the last session t, semantic features of testing samples $X\{x_1^t, x_2^t, \ldots, x_n^t\}$ from all seen categories are extracted by the trained model ϕ^t. By comparing cosine distances between features of each sample to all classes prototypes from p_1 to p_{c^t}, predictions can be made by assigning a testing sample x_n^t to a specific class i with the closest distance among all compared prototypes. Algorithm 1 depicts the learning process of our framework, where the γ of the total loss $\mathcal{L}_{\text{total}}$ is used to balance the model's plasticity for learning new concepts and stability for knowledge preservation.

Algorithm 1 Learning Process of CPL Framework

Input: A CNN-based feature extractor parameterized as ϕ. A sequence of labeled data $D^1, D^2, \ldots, D^t$

Output: A trained feature extractor ϕ^t, all saved prototypes $\{p_1, p_2, \ldots, p_i\}_{i=1}^{c^t}$

// Base session training on D^1, initialize base class weight w^b

for epoch n = 1, 2,... **do**

Compute $\mathcal{L}_{CE} = -y_i \log(p(\xi_i^1))$, $(x_i, y_i) \in D^1$, $\xi_i^1 \leftarrow \kappa\left(\phi^b(x_i)\right), w^b)$ calculated by Eq.2

Update w^b and ϕ^b through Backpropagation

end for

Save ϕ^b, $\{p_i\}_{i=1}^{c^1}$ and E^1 of D^1

//Incremental learning on $D^k, k \neq 1$

for $k \leftarrow 2$ to t sessions **do**

Combine exemplar set E^{k-1} and D^k as training data of session k.

for epoch n = 1, 2, ... **do**

Compute $\mathcal{L}_{total} = \gamma(\mathcal{L}_{PSS} + \mathcal{L}_{\text{PC}}) + \mathcal{L}_{PED}$

Update ϕ^k through Backpropagation

end for

Save ϕ^k and E^k by combining E^{k-1} with selected exemplars of $\mathcal{C}^k$ based on herding strategy, recalibrate $\{p_i\}_{i=1}^{c^k}$ by PR strategy

end for

D. Model Plasticity

Our method's plasticity for new knowledge is provided by PSS and PC losses.

1) *PSS:* In this part, the pairwise relationships of interclass and intraclass samples are exploited, followed by a detailed description of the PSS loss. With a randomly selected target as the compared anchor (the yellow dot pointed by a blue arrow), the structure (SSIM) and feature similarity scores of intraclass (yellow dots) and interclass (black dots) instances are provided in Fig. 6(a) and (b), respectively. It could be seen that both of the two scores for intraclass samples are higher

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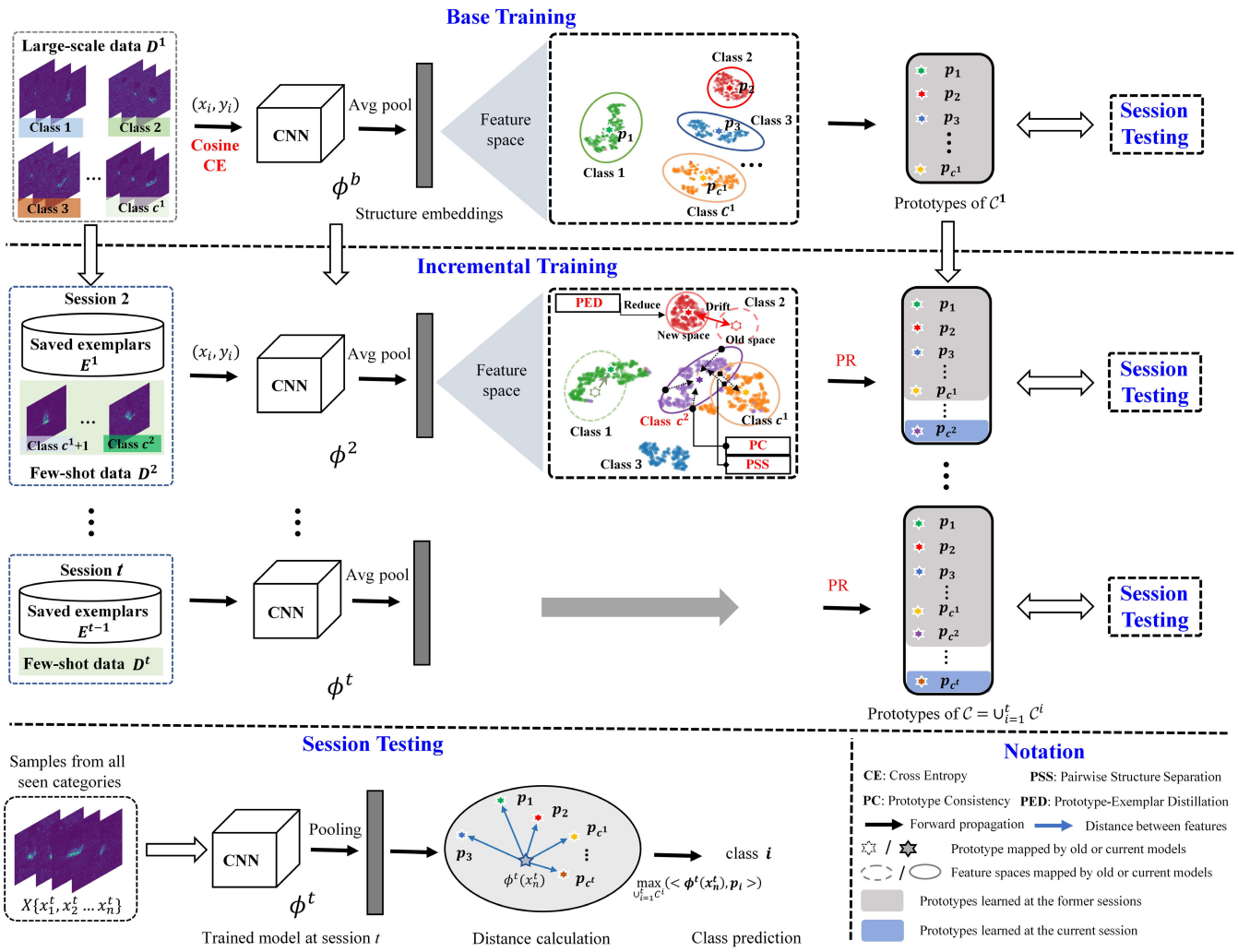


Fig. 5. Overall framework of CPL framework.

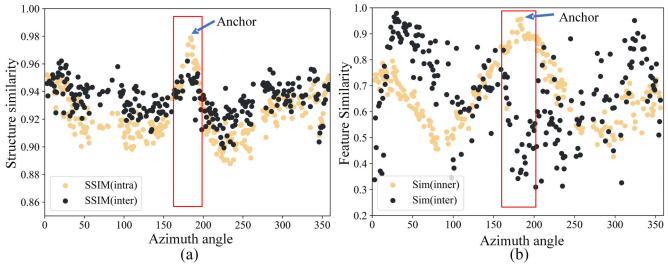


Fig. 6. Illustration of interclass similarity and intraclass discrepancy remained in target structures and high-dimensional features at specific azimuth regions. Scores are obtained by comparing a random anchor to the same class samples and those of another class. (a) Structure similarity (SSIM). (b) Feature similarity.

than those for interclass ones at a small region of target azimuths (red rectangles) but lower at other positions. Besides, a serious fluctuation exists in the two scores for intraclass pairs, as target azimuth changes, e.g., from 1.0 to 0.90 or to 0.30 for structure or feature similarity. Therefore, it is essential to enlarge discrepancies of interclass target features at near azimuths (high similarity scores) while making features of intraclass targets at different azimuths (low similarity scores) more compact. Illustrated in Fig. 7, a PSS loss is introduced and formulated as follows:

$$\mathcal{L}_{\mathrm{PSS}} = \frac{1}{N} \sum_{x_j}^{E^{t-1} \cup D^t} \sum_{x_i, x_i^+}^{\mathcal{D}^t} \max(m - d(x_i, x_i^+) + d(x_i, x_j^-), 0)$$

where

$$y_i \neq y_j, \quad d\left(x_i, x_{i/j}^{+/-}\right) = \left\langle \phi^t(x_i), \phi^t\left(x_{i/j}^{+/-}\right) \right\rangle \tag{3}$$

where x_i and x_i^+ are a selected anchor and a positive sample of class $y_i \in \mathcal{C}^t$ $(t > 1)$, respectively. x_j^- is a negative sample of class y_j selected from E^{t-1} and D^t. The positive and the negative samples are with different labels, i.e., $y_i \neq y_j$. N is the number of compared pairs. For a specific anchor, the same class samples with the top-N farthest cosine distances to the anchor are selected as positive instances. In contrast, samples of other classes with the top-N closest distances to the anchor are regarded as negative ones. Compared with triplet loss [49], where anchors are from all samples with compared pairs selected by Euclidean distance,

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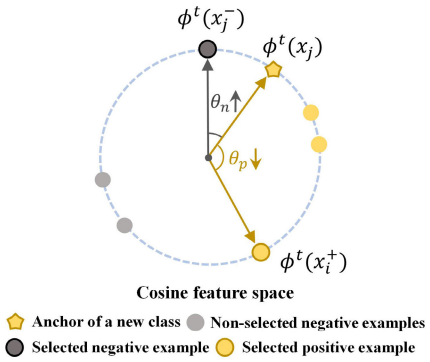


Fig. 7. Illustration of PSS loss.

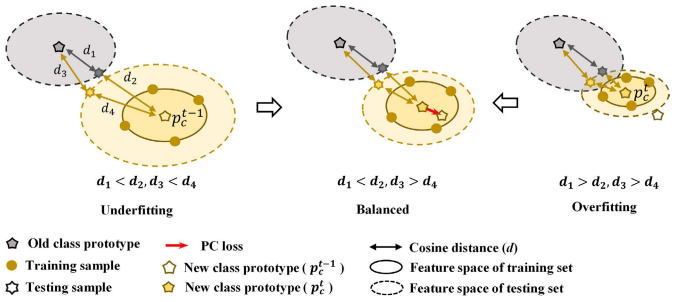


Fig. 8. Illustration of PC loss.

samples in D^t are selected as anchors in PSS loss to avoid inappropriate adaptions during learning from few-shot samples. Besides, the cosine criterion is employed to compute d considering its superiority of representing target azimuth-aware structures.

2) *PC:* In a sequential manner, the model's feature spaces evolve continually as Fig. 8. In the left subfigure, the initial model could maintain a compact low-dimensional space on the old class (gray dotted ellipse) but an underfitting one on the new (yellow dotted ellipse), which easily induces misclassification on boundary samples of the new class (the yellow-filled hexagon), e.g., $d_3 < d_4$. Conversely, in the right subfigure, the model trained on few-shot novel samples is overfitting to the new class, which possesses a biased and distorted feature space on both old and new classes, leading to a weak ability to discriminate the old class, e.g., $d_1 > d_2$. Inspired by the generalization of class prototypes, a PC loss is proposed to reach a balance. As formulated by Eq. (4), knowledge of new classes at the session t can be learned by our method without losing generalization by minimizing cosine distances between p_c^t and p_c^{t-1}. The p_c^t calculated by (5) is the prototype of class c produced by the current trainable model ϕ^t. The p_c^{t-1} is obtained by replacing ϕ^t with the fixed ϕ^{t-1}. As diagrammed in the middle subfigure of Fig. 8, a compact and general feature space of the new class is constructed

$$\mathcal{L}_{\text{PC}} = 1 - \frac{1}{|\mathcal{C}^t|}\sum_{c}^{\mathcal{C}^t}\left\langle p_c^t, p_c^{t-1}\right\rangle \tag{4}$$

$$p_c^t = \frac{1}{N_c}\sum_{i}^{\mathcal{D}^t}\mathbb{1}(y_i = c)\phi^t(x_i), \quad c \in \mathcal{C}^t \tag{5}$$

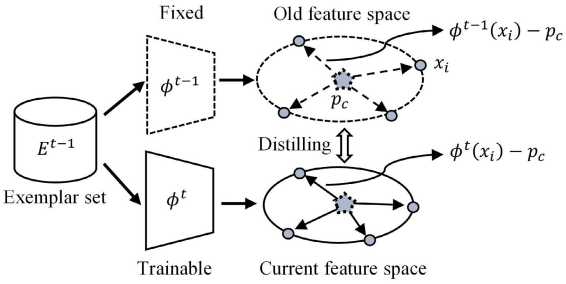


Fig. 9. Illustration of PED loss.

E. Model Stability

A PED loss and a PR strategy are proposed to overcome the catastrophic forgetting problem in the CPL.

1) *PED:* KD has been broadly used in most IL tasks to tackle the catastrophic forgetting issue. However, large-scale training data are required, which is not accessible in the FSCIL of SAR ATR. In contrast to distilling individual points at feature spaces, a PED loss is designed to provide rich and diverse supervision signals by regularizing the stability of cosine semantic structures spanned by limited exemplars and the corresponding prototypes. As formulated in (6), at session $t\,(t > 1)$, $\phi^{t-1}(x_i)$ and $\phi^t(x_i)$ are semantic features of a sample x_i of class y_i from the saved exemplar set E^{t-1} at session t. p_c is the prototype of the old class c. The indicator function $\mathbb{1}(y_i = c)$ returns 1 if $y_i = c$ else 0. N is the number of samples in E^{t-1}. By minimizing (6), internal semantic structures spanned by class-aware exemplars and prototypes can be aligned at learning sessions. Intuitively, we diagram PED loss in Fig. 9

$$\mathcal{L}_{\text{PED}} = -\frac{1}{N}\sum_{i}^{E^{t-1}}\sum_{c}^{\cup_{i=1}^{t-1}\mathcal{C}^i}\mathbb{1}(y_i = c) \times \langle \phi^{t-1}(x_i) - p_c, \phi^t(x_i) - p_c\rangle. \tag{6}$$

Notably, different from distilling semantic shifts of isolated points at feature spaces [36], [37], internal class-aware structures are mainly considered by PED loss, considering the abundant structural cues represented as semantic links between class-aware exemplars and prototypes.

2) *PR:* Although the learned knowledge can be preserved by PED loss, semantic structures of old classes inevitably suffer from drifts due to the scarcity of supervision provided by limited exemplars, which may lead to a serious PD on old classes based on the previous prototypes. We aim to mitigate the drifts and propose a simple but effective postprocessing strategy called PR, which is formulated as (7). At session t, p_c^{t-1} is the prototype of class c before calibration. p_c^t is the calibrated prototype. If c is an old class, i.e., $c \notin \mathcal{C}^t$, the prototype p_c^t is rectified by compensating p_c^{t-1} with a class-aware offset δ_c^t. Since the overall samples of old classes are unavailable, we estimate the offset δ_c^t by (8), where the average semantic offsets of

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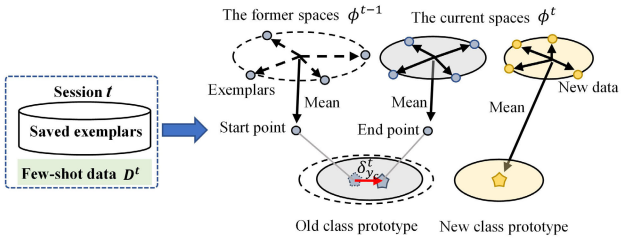


Fig. 10. Illustration of PR strategy.

class-specific exemplars E_c produced by $\phi^{t-1}(x_i)$ and $\phi^t(x_i)$ are adopted. $|E_c|$ is the number of exemplars of class c. The indicator function $\mathbb{1}(y_i = c)$ returns 1 if $y_i = c$ else 0. As diagrammed in Fig. 10, the ultimate location of an old class prototype (the real gray pentagon) is obtained by shifting the original (the gray dotted pentagon) with an offset pointed by the red arrow. The start and end points of the arrow refer to the mean features of old-class exemplars projected by the ϕ^{t-1} and ϕ^t, respectively. The novel class prototype (the yellow pentagon) is the mean of target features at current spaces. Notably, the network is fixed during calibration, and the post-calibrated prototypes are used for prediction at the current session

$$p_c^t = \begin{cases} p_c^{t-1} + \delta_c^t, & c \notin \mathcal{C}^t \\ p_c^t \end{cases} \tag{7}$$

$$\delta_c^t = \frac{1}{|E_c|} \sum_i \mathbb{1}(y_i = c)(\phi^t(x_i) - \phi^{t-1}(x_i)). \tag{8}$$

F. NCM Classifier

Due to the scarcity of significant information provided by few-shot novel samples, learning discerning features of new classes by a parametric classifier is intractable. Also, the imbalanced training samples provided by the base and the incremental sessions inevitably induce a biased representation of different classes by a trained CNN model. In contrast to the vanilla NCM classifiers [36], in which Euclidean criterion is used for evaluation, a cosine-based NCM classifier is leveraged in our method for balanced inference, considering the form of cosine criterion and its advantages of representing target structural details without losing generalization. As formulated by (9), $\phi^t(x_i)$ is the output of the trained model ϕ^t on a testing sample x_i. p_c represents post-calibrated prototypes of all seen classes until session t. The prediction y_i^* can be obtained by assigning x_i to the class c, of which cosine similarity to the p_c is the highest among all prototypes

$$y_i^* = \arg\max \langle \phi^t(x_i), p_c \rangle, \quad c = 1, 2, \ldots, c^t. \tag{9}$$

IV. Experiments

In this section, we first detail the used dataset, training settings, and evaluation criteria. Ablation studies show the contributions of losses and strategies of our method. Then, the performance of our method compared with other state-of-the-art methods at various FSCIL settings is provided. Finally, the effects of hyperparameters are discussed.

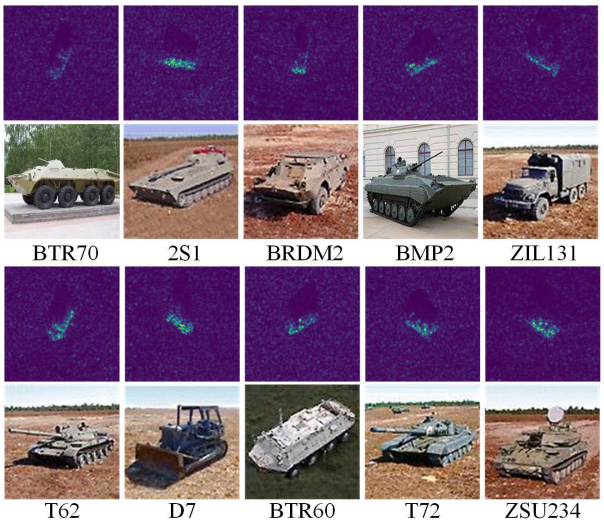


Fig. 11. Target instances of shuffled classes under the SOC of the MSTAR dataset in SAR and optical images. Targets in SAR images are plotted with pseudo-colors.

A. Dataset Description

In experiments, the MSTAR dataset [50], sponsored by the Defense Advanced Researched Projects Agency and the Air Force Research Laboratory, is used to evaluate the performance of compared methods considering its completeness, diversity, and standardization. It contains targets imaged at various configurations, such as types, azimuths, depression angles of radar, and so on. The public part of the dataset includes ten types of Soviet armored ground vehicles imaged at X-band spotlight mode with 0.3-m resolution. Specifically, targets under the SOC are imaged at 17° and 15° depression angles and cover azimuths ranging from 0° to 360°.

To formulate a real-world learning scenario, we construct a dataset for FSCIL of SAR ATR from the SOC of the MSTAR. Following the building process of optical FSCIL datasets in topology-preserving knowledge incrementer (TOPIC) [16], we first shuffle the ten classes randomly. Fig. 11 shows targets of shuffled classes in SAR and optical images. The first four classes, i.e., bronetransporter-70 (BTR70), 2S1, BRDM2, and BMP2, serve as the base session classes to imitate pre-collected knowledge in cooperative situations. The training samples of these classes are imaged at the 17° depression angle with full azimuths. The remaining six classes are used for the later incremental sessions, which can be regarded as new concepts collected in noncooperative situations. Each incremental session's training data are constructed by randomly picking a few samples (five shots in experiments) per class from the original samples imaged at the 17° depression angle. For the test set, targets of these classes imaged at 15° depression angle with all azimuths are used for evaluation. For a stable judgment, training instances of each incremental class are sampled $T = 10$ times, and the final performance is obtained by averaging all testing results. Tables I and II detail the target configurations of the constructed dataset.

B. Training Settings

In experiments, only two exemplars are selected by the herding strategy [36] for knowledge replay in our method and

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TABLE I
LARGE-SCALE DATA OF BASE SESSION

Class order	Class name	Serial No.	Training set		Testing set	
			Depression	Number	Depression	Number
1	BTR70	c71	17°	233	15°	196
2	2S1	b01	17°	299	15°	274
3	BRDM2	E-71	17°	298	15°	274
4	BMP2	9563	17°	233	15°	196

TABLE II
FEW-SHOT DATA OF INCREMENTAL SESSIONS

Class order	Class name	Serial No.	Training set		Testing set	
			Depression	Number	Depression	Number
5	ZIL131	E12	17°	5	15°	274
6	T62	A51	17°	5	15°	273
7	D7	92v13015	17°	5	15°	274
8	BTR60	k10yt7532	17°	5	15°	195
9	T72	132	17°	5	15°	196
10	ZSU234	d08	17°	5	15°	274

the compared exemplar-based methods. The balanced factor γ in $\mathcal{L}_{\text{total}}$ is set to 1.0. The m and N in PSS loss are set to 0.5 and 1, respectively. ResNet-18 pretrained on the ImageNet [51], a lightweight deep residual network of the ResNet series [52], serves as the feature extractor ϕ in the CPL and all compared methods to obtain a considerable cornerstone for network adaption without losing generalization. We remove the original linear classifier to satisfy dynamic class numbers at different learning sessions. Other weights in compared methods, such as nodes of linear classifiers in incremental classifier and representation learning (iCaRL), learning a unified classifier incrementally via rebalancing (LUCIR), and incremental DL vector quantization classification (IDLVQC), are initialized following the original papers. At the base session, the compared methods are trained by stochastic gradient descent (SGD) optimizer for 50 epochs with a mini-batch size of 32. The momentum and weight decay are set to 0.9 and $5e-4$, respectively. The initial learning rate is $1e-2$ and is decayed at epochs of 30 and 40 with a rate of 0.1. The same optimizer and configurations are used for training the models at incremental sessions except for the initial learning rate, which is reset to $1e-4$ and is decayed with the rate of 0.1 at steps of both 30 and 60. Images are normalized with ranges from 0 to 1 and cropped into sizes of 64×64 pixels. The training samples are randomly rotated within a small degree (15°) for data augmentation to maintain the diversity. All experiments are conducted on a server equipped with Nvidia GTX 3090 graphic cards and CUDA 11.1 support.

C. Evaluation Protocols

Three protocols are adopted for judgment. To evaluate successive performance, we calculate the classification accuracy of compared methods at each session. Besides, the average accuracy (Avg.Acc) is provided to judge the holistic performance of methods by averaging successive results until the current session. The PD rate [19] measures the absolute PD of models at the last session, which is calculated by subtracting the accuracy of the last session from that of the base session. The larger the values of the PD, the more serious the forgetting issue of compared methods.

D. Benchmark Descriptions

In experiments, our method is compared with several task-specific approaches. For traditional learning, a CNN-based classifier is directly fine-tuned on samples of novel classes at the current session by the current session and is notated as a fine-tuned CNN (Ft-CNN). Joint-CNN trains all classes in an offline manner by using all training data until the current session. Classifier retraining (cRT) [53] is a rebalanced classifier for long-tailed recognition. For IL, iCaRL [36] preserves old knowledge by distilling old knowledge from exemplars selected by a herding strategy and predicts targets by the Euclidean-based NCM classifier. LUCIR [37] introduces cosine normalization and two regularization losses on the old and the novel target features to tackle imbalanced class magnitudes during IL. For few-shot learning (FSL), ProtoNet [54] learns a metric space by episodic training and performs classification through computing distances of query samples to the prototypical representation of each class. RelationNet [55] learns a relation module by episodic training and classifies targets by computing relation scores along feature channels between query images and class prototypes supported by limited examples. For FSCIL, TOPIC [16] first formulates and solves the problem by preserving knowledge through a neural gas network [56] by maintaining the topology of features of different classes. IDLVQC [40] introduces several trainable vectors to capture representative features of novel classes. Finding flat minima (F2M) [43] overcomes the catastrophic forgetting problem by finding the local minima and acquires an incremental fully connected classifier by constraining ranges of trainable weights. CEC [19] proposes a continually evolved classifier to adaptively adjust class weights and features of testing samples coherently by a graph attention network. Forward compatible training (FACT) [57] achieves state-of-the-art performance by designing a forward-compatible model in which virtual prototypes are assigned to squeeze the embedding of known classes and reserve for new ones.

E. Ablation Studies

1) Effects of Cosine Criterion: In this section, the effects of the cosine criterion for representing significant features of targets are investigated. To purely verify the contributions, we consider an intransigent model that does not adapt to new tasks. Most specifically, the model is solely trained on the base session data D^1 by the three criteria, i.e., FC, Euclid, and cosine, and is directly acted as the feature extractor ϕ^1 for inference at the later sessions by the NCM classifier. In particular, the model is first trained and tested by the same criteria and then all by cosine criterion.

1) *Quantitative Analyses:* Table III shows the performance of models trained and tested by different criteria. Criteria used at the training and the testing stages are placed on the left-hand and the right-hand sides of $\rightarrow$, respectively. From the first two rows of Table III, we may conclude that only considering target amplitude features within the Euclid criterion can easily lead to inaccurate predictions due to serious background clutters. However, by exploring relationships of target features by

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TABLE III
SUPERIORITY OF USING COSINE PROTOTYPES FOR FSCIL OF SAR ATR UNDER ONE-*Way* FIVE-*Shot* SETTING

Methods	Sessions (%)							Avg. Acc	PD ↓
	1	2	3	4	5	6	7		
Euclid→Euclid	82.85	66.53	55.41	49.80	46.15	43.43	38.31	54.64	44.54
FC→FC	97.98	75.91	61.97	52.84	47.57	43.82	39.27	59.89	58.71
Euclid→Cosine	94.14	79.62	67.13	61.97	53.39	46.02	41.15	63.35	52.99
FC→Cosine	98.62	87.80	75.98	65.68	63.32	60.20	53.53	72.16	45.09
Cosine→Cosine	99.89	88.29	73.82	68.07	65.68	61.69	57.03	73.49	42.86

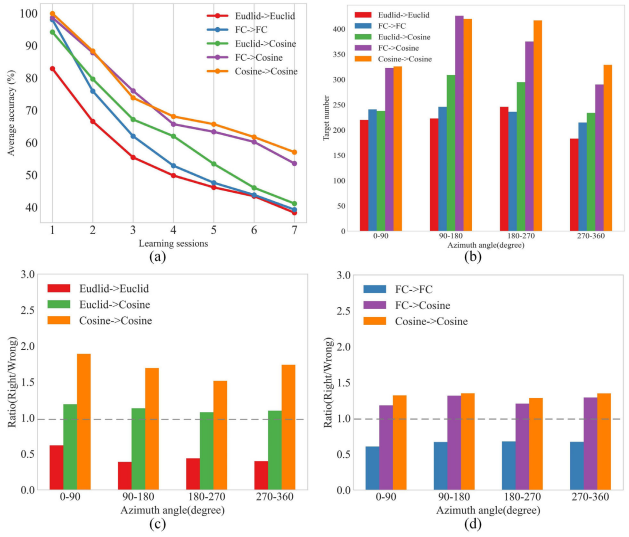


Fig. 12. Visualized contributions of cosine criterion for FSCIL in SAR images at each session and different azimuth intervals. (a) Performance curves of Avg.Acc. (b) Numbers of correctly predicted targets at different intervals of azimuths. (c) Ratio r under Euclid criterion. (d) Ratio r under FC criterion.

FC criterion, target discriminative characteristics can be captured, and the model's performance can be improved to some extent. From the third and the fourth rows of Table III, structural cues of targets at Euclid and FC feature spaces can be further exposed and utilized at cosine-based feature space, significantly improving the performance of the original models for classifying both base and incremental classes. Notably, maybe because of intrinsic information of targets captured by the feature space learned by the FC, the accuracy of the model can be further boosted when judging by the cosine criterion. Furthermore, as shown in the last column of Table III, the model trained and tested by cosine criterion performs the best among all these comparisons benefiting from the superiority of capturing normalized structural features of targets as well as being robust to background interference.

2) *Qualitative Analyses:* Visualized performance of models trained and tested by different criteria is also detailed in Fig. 12. Performance curves of models judged by cosine criterion are steadily superior to those tested by Euclid and FC criteria at most learning sessions. Besides, the numbers of correctly classified targets by these models at different azimuth intervals are shown in Fig. 12(b). Noticeably, the numbers of targets predicted by models with Euclid and FC criteria at both stages are almost steady and significantly lower than those predicted by the same models with the cosine at the testing stage. Furthermore, target details can be further captured by the model trained by cosine criterion, leading to a distinct increase in the number of rightly predicted targets, especially at the two intervals, i.e., 90 to 180 and 180 to 270. Besides, the superiority of the cosine criterion for rectifying wrong predictions under the FC and the Euclid criteria is explored. We define the ratio (r) of $\kappa\left(\phi(x_{\text{wrong}}), \mathbf{v}_{\text{right}}\right)$ to $\kappa\left(\phi(x_{\text{wrong}}), \mathbf{v}_{\text{wrong}}\right)$ in (10) to quantify the distance from the misclassified samples to different types of prototypes. x_{wrong} comes from testing samples misclassified by Euclid or FC criterion. $\mathbf{v}_{\text{right}}$ and $\mathbf{v}_{\text{wrong}}$ are prototypes of ground-truth (right) and misclassified (wrong) labels, respectively. The larger the value of r, the more accurate the representation of ground-truth (right) prototypes to misclassified targets

$$r = \frac{\kappa\left(\phi(x_{\text{wrong}}), \mathbf{v}_{\text{right}}\right)}{\kappa\left(\phi(x_{\text{wrong}}), \mathbf{v}_{\text{wrong}}\right)}. \tag{10}$$

From Fig. 12(c) and (d), the ratios obtained by models trained and tested under Euclid and FC criteria are lower than 1.0 at all azimuth intervals, demonstrating a serious misalignment between target features and their prototypes. However, the same ratio is higher than 1.0 when judging the misclassified targets by the two types of prototypes under the cosine criterion (see the green and purple bars for examples). Furthermore, the ratio can be further boosted when judging these targets by the model learned and tested by the cosine criterion (the yellow bar), suggesting the effectiveness of prototypes for representing target characteristics under the cosine criterion. In addition, Fig. 13 shows the Grad-CAM [58] of target features at various azimuths. It could be seen that models learned by Euclid and FC criteria pay less attention to significant components and contours of targets due to target discrete backscattering parts and background interference. However, target components with salient scattering signatures (see the last three columns for examples) are captured and highlighted by models tested under cosine criterion, verifying its effectiveness for capturing target significant and detailed cues.

2) Effects of PSS Loss: The contributions of PSS loss for enhancing the model's plasticity are exploited in this section. Numerically, as shown in the first three and the fifth rows of Table IV, our method with PSS loss steadily performs better than that trained by cosine CE loss (the first row of Table IV)

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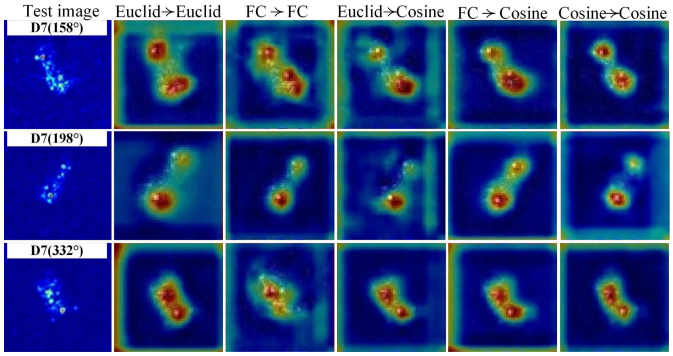


Fig. 13. Grad-CAM of targets semantic features mapped by the model trained and tested under different criteria.

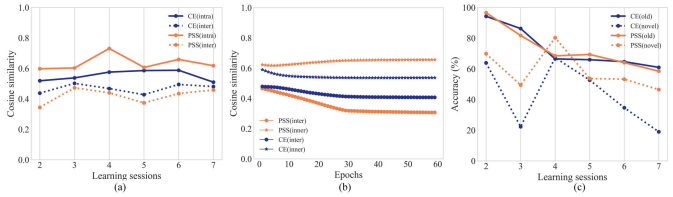


Fig. 14. Visualized contributions of PSS loss at each session and overall learning process. (a) Cosine similarity of intraclass novel features and features between the novel and old classes at incremental sessions. (b) Cosine similarity curves of novel target features and interclass features extracted by different models during the learning process. (c) Classification accuracy of different models for novel and old classes at incremental sessions.

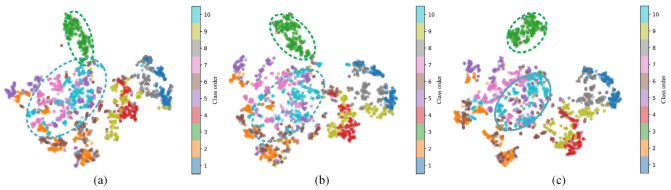


Fig. 15. t-SNE of feature spaces learned at the final session. Feature points of a base class and a new class are encompassed by the green and the blue dotted circles, respectively. (a) Feature spaces before adaption. (b) Feature spaces after adapted by a cosine CE loss. (c) Feature spaces after adapted by the PSS loss.

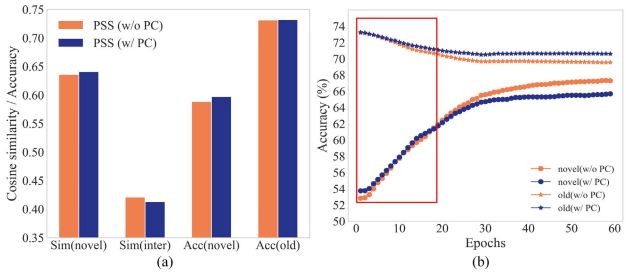


Fig. 16. Visualized contributions of PC loss. (a) Cosine similarity of novel and interclass targets and averaged classification accuracy of new and old classes at all incremental sessions. (b) Statistic trend of classification accuracy of our method (w/ or w/o PC loss) for both old and new classes during network optimization.

and even those learned by CE and other proposed strategies (see the second and third rows of Table IV for examples), suggesting the effectiveness of PSS loss for learning new concepts. Specifically, the Avg.Acc of the method with PSS loss is 1.36% and 0.75% higher than scores achieved by the method with only CE and that with the three losses (CE, PED, and PR), respectively. Besides, the PD of the method with PSS loss is 0.88% and 0.14% lower than those achieved in the same models, respectively.

Apart from these comparisons, qualitative results are also shown in Fig. 14. First, more compact and discriminative features of novel classes can be learned and maintained by the model trained by PSS loss than by CE loss at all sessions. Specifically, for intraclass cosine similarity of novel classes, scores of the method with PSS loss [the solid line in orange and the line marked with orange ⋆ in Fig. 14(a) and (b)] are steadily higher than those of the method with CE loss (the same lines in dark blue). Second, discrepancies among interclass features extracted by the model with PSS loss are much more distinct than those extracted by the model with CE loss. For example, the results of the method with PSS loss plotted as the dotted curve in orange and the curve marked with orange • in Fig. 14(a) and (b) are lower than those of the method with CE loss (the same curves in dark blue). Third, the method with PSS loss outperforms that with CE loss on classification accuracy of novel classes by a large margin in most sessions. In Fig. 14(c), the dotted line in orange is higher than that in dark blue. Intuitively, the t-distributed stochastic neighbor embedding (t-SNE) of feature spaces is shown in Fig. 15. It can be noticed that features of the new class (scatters in light blue color) learned by the model with PSS loss take up a smaller region than the models before adaption or learned by cosine CE loss. Besides, points of base classes mapped by the model with PSS loss (scatters in green color) are more compact than those projected by the other two models, suggesting the effectiveness of PSS loss for representing target informative cues.

3) Effects of PC Loss: The contributions of PC loss are investigated in this part. Numerically, compared with the method with PSS loss (the fifth row of Table IV), the performance of our method with both PSS and PC losses, shown in the sixth row of Table IV, can be further improved, which is 0.17% higher and 0.3% lower than the method with PSS loss on the Avg.Acc and the PD, respectively. Also, the performance of our method with the two losses is improved at most incremental sessions. The qualitative results are shown in Fig. 16. Based on the first two columns of Fig. 16(a), cosine similarity of intraclass novel targets and the interclass ones can be improved and reduced at all incremental sessions, respectively, because of a special regularization on feature consistency between target features and their prototypes by PC loss. Correspondingly, the accuracy of our method for classifying novel classes can be slightly boosted without losing performance on the old. Fig. 16(b) shows the accuracy trends of old and novel classes during IL. At the beginning and the initial period of learning, our method with PC loss achieves a higher starting point on learning new-class knowledge without a serious PD on the old compared with the method without PC loss. Although the method without PC loss performs better on classifying novel classes as the later epochs, the performance on old classes encounters an apparent decrease resulting from ambiguous features between novel and old classes.

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TABLE IV

EFFECTIVENESS OF COMPONENTS WITHIN THE PROPOSED FRAMEWORK TESTED ON THE MSTAR DATASET UNDER ONE-*Way* FIVE-*Shot* SETTING

Plasticity			Stability			Sessions (%)							Avg. Acc	PD ↓
CE	PSS	PC	PED	PR	NCM	1	2	3	4	5	6	7		
✓						99.89	89.84	74.56	66.66	66.00	61.99	56.27	73.60	43.62
✓			✓			99.89	89.85	74.91	68.08	66.91	62.07	57.29	74.14	42.60
✓			✓	✓		99.89	89.85	74.97	68.28	67.18	62.30	57.01	74.21	42.88
✓			✓	✓	✓	99.89	89.83	75.88	71.42	68.76	64.26	58.20	75.46	41.69
	✓				✓	99.89	90.55	75.83	70.30	67.86	63.14	57.15	74.96	42.74
	✓	✓			✓	99.89	90.59	75.76	70.76	67.96	63.52	57.45	75.13	42.44
	✓	✓	✓		✓	99.89	90.71	76.28	71.10	68.46	64.38	57.83	75.52	42.06
	✓	✓	✓	✓	✓	99.89	90.60	76.96	72.92	70.39	66.71	60.36	76.83	39.53

4) Effects of PED Loss: The model's catastrophic forgetting of the old knowledge is mitigated by PED loss of the CPL. As shown in the seventh row of Table IV, our method with PED loss performs better than that without the loss at the final and intermediate sessions. Specifically, the Avg.Acc and the PD of our method with the loss are 75.52% and 42.05%, which are 0.39% higher and 0.38% lower than those of the method without the loss, respectively. To further verify its effectiveness, the Kullback–Leibler (K–L) divergence is employed to quantify the ability at both feature and logit levels. As formulated by (11), O^{t-1} and O^t are activations of class-aware testing samples predicted by models at session $t-1$ and t, respectively. x_i is a sample of testing set S^{t-1} at session $t-1$

$$D_{K-L}(O^{t-1}||O^t) = \sum_{i}^{S^{t-1}} o^{t-1}(x_i) \log\left(\frac{o^{t-1}(x_i)}{o^t(x_i)}\right). \tag{11}$$

At feature level, $o^{t-1}(x_i)$ and $o^t(x_i)$ are target features extracted by ϕ^{t-1} and ϕ^t, respectively. The $D_{K-L}(O^{t-1}||O^t)$ directly measures the forgetting degree of the CPL for old class knowledge, and the results of incremental sessions are shown in Fig. 17(a). At logit level, the $o^{t-1}(x_i)$ is calculated as $p_j^{\mathsf{T}}\phi^{t-1}(x_i)$, $\{p_j\}_{j=1}^{c^{t-1}}$. $\phi^{t-1}(x_i)$ is the same as that at feature level. Hence, the $D_{K-L}(O^{t-1}||O^t)$ measures the degree of misalignment between internal structures of old classes and their prototypes. The results are shown in Fig. 17(b). Some conclusions can be summarized as follows. First, scores of the two levels of divergence decrease significantly at all incremental sessions by using PED loss. Because of the distillation of correlations between selected exemplars and original prototypes, both overall and intrinsic semantic structures of old class characteristics can be well preserved, alleviating distortion of old feature spaces projected by the current model. Second, degrees of overcoming forgetting the old knowledge evaluated at the two levels (the residual values between orange and dark blue bars at each session) are different among learning sessions. Due to certain confusion on configurations and functions among different classes, feature spaces of old classes inevitably distort once adapted to new knowledge, especially when some novel classes are highly similar to the old. Nonetheless, our method with PED loss still performs better than that without the loss, verifying its advantage of remembering representative characteristics of old concepts.

5) Effects of PR Strategy: In the CPL, the misalignment between prototypes and old-class feature spaces is alleviated by PR strategy. As shown in the penultimate row of Table IV, a significant improvement in classification accuracy at all incremental sessions can be achieved by the method with PR compared with the method without the strategy. Specifically, the Avg.Acc and the PD of our method with PR strategy are 76.83% and 39.53%, which are 1.31% higher and 2.53% lower than those without the strategy, respectively. Besides, our method's advancement of classifying all seen classes becomes more prominent, as the session increases, verifying the contribution of PR strategy for correcting accumulation drifts of feature spaces at successive sessions.

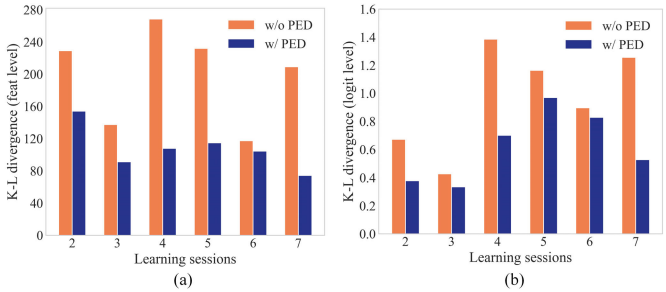


Fig. 17. Illustration of alleviating the catastrophic forgetting issue by PED loss. K–L divergence between old-class knowledge processed by the former fixed and that by the current trainable models at each incremental session. (a) K–L divergence at the feature level. (b) K–L divergence at the logit level.

Furthermore, we also quantify the superiority as K–L divergence at the logit level depicted in Section IV-E4. For the D_{K-L} of model without PR at current session t, $o^{t-1}(x_i)$ and $o^t(x_i)$ are calculated as $p_j^{\mathsf{T}}\phi^{t-1}(x_i)$ and $p_j^{\mathsf{T}}\phi^t(x_i)$, respectively. However, for the D_{K-L} of the method with PR, $o^{t-1}(x_i)$ and $o^t(x_i)$ are calculated as $p_j^{\mathsf{T}}\phi^{t-1}(x_i)$ and $\hat{p}_j^{\mathsf{T}}\phi^t(x_i)$, respectively. Here, $\forall p_j \in \{p_j\}_{j=1}^{C^{t-1}}$ is a class prototype learned at session $t-1$, and $\forall \hat{p}_j \in \{\hat{p}_j\}_{j=1}^{C^{t-1}}$ is the post-calibrated prototype of the same class after learning of session t. We calculate D_{K-L} of both the former and base classes, and the results are shown in Fig. 18(a) and (b), respectively. The two criteria measure distortion degrees of the former class and base class feature spaces at the current session, respectively. Some conclusions can be summarized as follows. First, the misalignment between prototypes and features of classes learned at the $t-1$ session can be alleviated after calibration at session t, improving our method's ability to capture discriminative features of old knowledge. In Fig. 18(a), the values of D_{K-L} of the method with PR significantly decrease compared with those of the method without PR. Second, In Fig. 18(b), values

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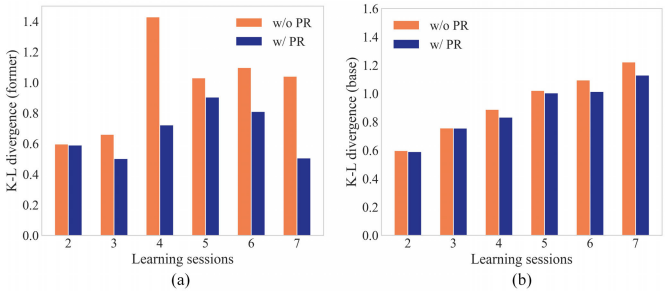


Fig. 18. Illustration of alleviating the misalignment issue by PC strategy. K–L divergence of logits of old and base classes at each incremental session. (a) K–L divergence of old-class logits projected by the original and post-calibrated prototypes. (b) K–L divergence of base-class logits projected by the original and post-calibrated prototypes of base classes.

of the two methods all come up, as the session increases. As new knowledge is learned in a stream format, feature spaces of base classes are continually compressed, leading to a consistent decrease in models' performance for classifying base classes. However, our method with PR strategy is inferior to those achieved by the method without PR, verifying the contribution of PR strategy for rectifying the misalignment of intrinsic structures of old classes at successive sessions.

F. Benchmark Comparison

In this section, extensive experiments are conducted to evaluate our method.

1) Numerical Results and Performance Curves: Our method is compared with various benchmarks at three FSCIL settings of SAR ATR tasks, including one-*way* five-*shot*, two-*way* five-*shot*, and one-step five-*shot*. The numerical results are given by Tables V–VII, respectively. Fig. 19 shows the performance curves. Shortly, some conclusions are summarized as follows.

1) Holistically, the top performance is achieved by our method compared with these methods except for Joint-CNN and cRT at the three FSCIL settings. Undoubtedly, benefiting from acquiring all training samples of both old and novel classes and sophisticated long-tailed training strategies, competitive performance is achieved by Joint-CNN and cRT. Nevertheless, our method still achieves near and even higher results than the scores of the two methods (see the red point at the second session of one-way five-shot setting for an example). Apart from this, our method significantly outperforms most methods designed for the IL, FSL, and FSCIL tasks by a large margin on the Avg.Acc and the PD. Specifically, Avg.Acc of our method is 76.83%, 76.23%, and 79.16% at the three settings, respectively, which is 5.04%, 4.17%, and 4.89% superior to the second-ranked method (CEC). Although knowledge of old classes is preserved by the compared exemplar-based methods, such as iCaRL, LUCIR, and IDLVQC, the neglect of relationships between exemplars and prototypes and the vanilla classification strategies, such as Euclid in iCaRL and IDLVQC, can easily hinder them from preserving old knowledge and capturing discerning novel features accurately.
2) The long-term IL performance with a lower precision drop rate is obtained by our method at intermediate sessions, benefiting from the well-estimated prototypes and the well-separated features of all seen classes. From the three tables and Fig. 19, our method consistently achieves the best classification accuracy at all sessions compared with all these approaches except for the Joint-CNN and the cRT. Because of the particular considerations of decision boundaries and structural relationships among successive classes, clear separations among classes can be constructed in our method. In addition, benefiting from representative features captured by post-calibrated prototypes, the drifts of old features can be alleviated when learning novel classes, leading to more minor accumulated deviations than the compared methods.
3) More competitive performance is achieved by our method than compared approaches at the base session. Undoubtedly, benefiting from large-scale training samples, rich and discerning information of base classes can be fully captured by these methods. Analogously, the base session is similar to a traditional classification task. Features extracted by the base models are representative after a tedious optimization. Nevertheless, discrepancies still exist among different methods resulting from diversified classification criteria and training strategies. For example, compared with the Euclidean criterion used by iCaRL, F2M, and IDLVQC, relational characteristics are also considered by the linear classifiers in Ft-CNN, Joint-CNN, and cRT, leading to a stronger ability of these methods to capture multidimensional characteristics of targets.

2) Evolution of Feature Spaces: In addition, features of testing samples extracted by our method at each incremental session are also analyzed by t-SNE. The results for the three FSCIL settings of SAR ATR are shown in Figs. 20–22, respectively. Features of testing samples and class-aware prototypes are represented as colored points and numbers, respectively. Orders and colors corresponding to different classes are displayed on the bottom and the right sides of the figures, respectively. Specifically, feature spaces of novel classes at each incremental session are labeled with colored ellipses. We also show the t-SNE results of features projected by Ft-CNN in the last subfigure of Fig. 20. Some conclusions could be provided as follows.

1) Our method can distinctly separate interclass features and improve the compactness of intraclass samples by using few-shot samples. Compared with ambiguous features extracted by Ft-CNN, features of most novel classes mapped by our method are representative with clear boundaries. Obviously, benefiting from the particular separations of samples with similar structures and characteristics between the novel and all old samples, features of novel classes circled by colored ellipses at each incremental session are aggregated at most incremental sessions (see features of D7, BTR60, and

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TABLE V
COMPARISON WITH STATE OF THE ARTS UNDER ONE-*Way* FIVE-*Shot* SETTING ON THE MSTAR DATASET

Methods	Sessions (%)							Avg. Acc	PD ↓
	1	2	3	4	5	6	7		
Ft-CNN	98.94	76.21	62.01	52.02	47.55	43.08	38.12	59.70	60.82
Joint-CNN	98.94	85.05	77.17	78.30	73.86	70.34	67.10	78.68	31.84
cRT [53]	98.94	84.35	75.72	74.44	70.91	65.44	61.29	75.87	37.65
iCaRL [36]	92.12	80.37	69.35	61.16	58.02	53.47	47.75	66.03	44.37
LUCIR [37]	99.79	81.93	69.21	60.03	55.43	51.73	46.33	66.35	53.46
ProtoNet [54]	90.87	81.51	68.10	64.39	58.36	54.11	47.80	66.45	43.07
RelationNet [55]	80.56	63.85	51.83	47.15	42.37	38.43	34.03	51.17	46.53
TOPIC [16]	91.10	85.27	72.77	63.25	61.24	56.03	50.03	68.53	41.07
IDLVQC [40]	97.02	82.91	70.94	61.94	58.70	53.85	47.65	67.57	49.37
F2M [43]	85.30	74.06	63.16	56.82	50.87	46.83	41.55	59.79	43.75
CEC [19]	90.54	80.52	72.27	72.53	66.97	61.76	57.97	71.79	32.57
FACT [57]	98.85	87.36	67.14	64.24	49.50	46.69	45.01	65.54	53.84
Ours (CPL)	**99.89**	**90.60**	**76.96**	**72.92**	**70.39**	**66.71**	**60.36**	**76.83**	**39.53**

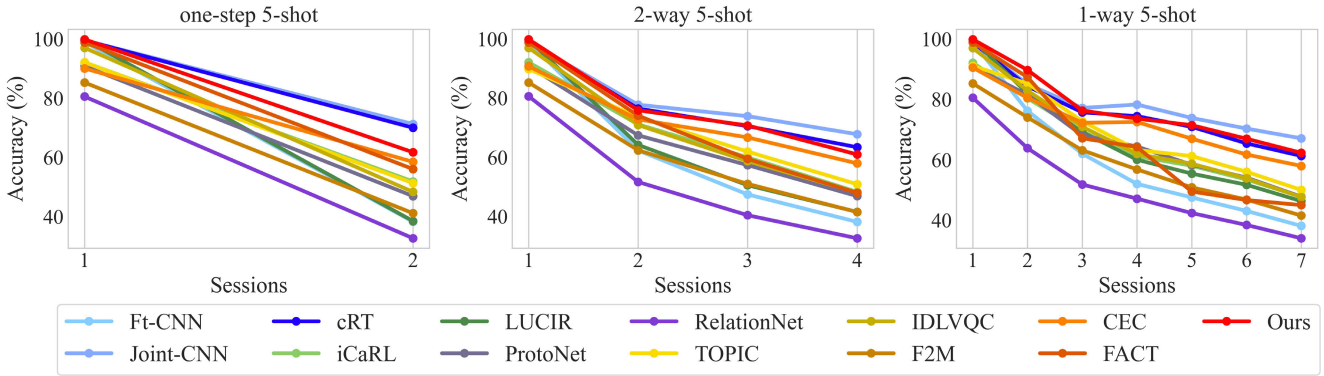


Fig. 19. Performance curves of compared methods under three FSCIL settings on the MSTAR dataset.

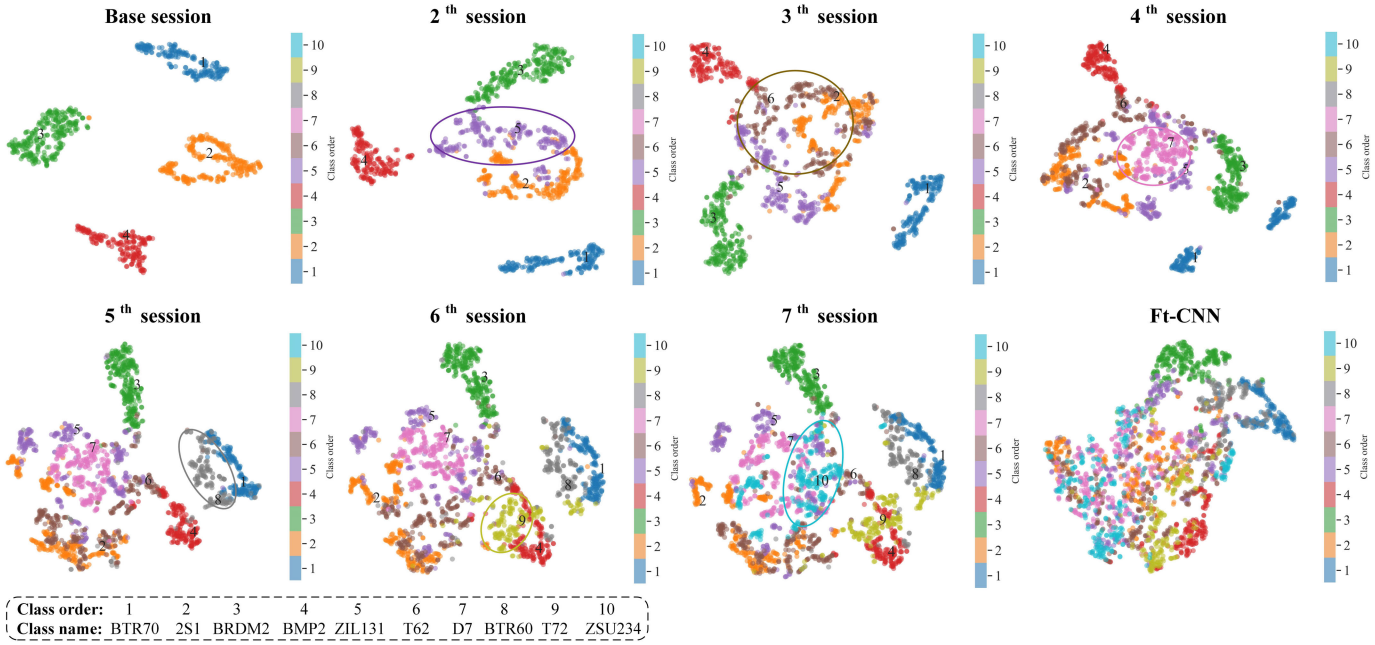


Fig. 20. t-SNE of ten-class semantic features under one-*way* five-*shot* setting.

T72 for examples in Fig. 20). Finally, features of all classes, shown in the penultimate subfigure of the three figures, are with distinct separations, verifying the ability of our method to learn novel knowledge from few-shot samples.

2) Topological structures of old classes mapped by our model are relatively stable across learning sessions, which suggests the mode's ability to preserve old knowledge well. According to the t-SNE results shown in Figs. 20 and 21, features of old classes mapped by our

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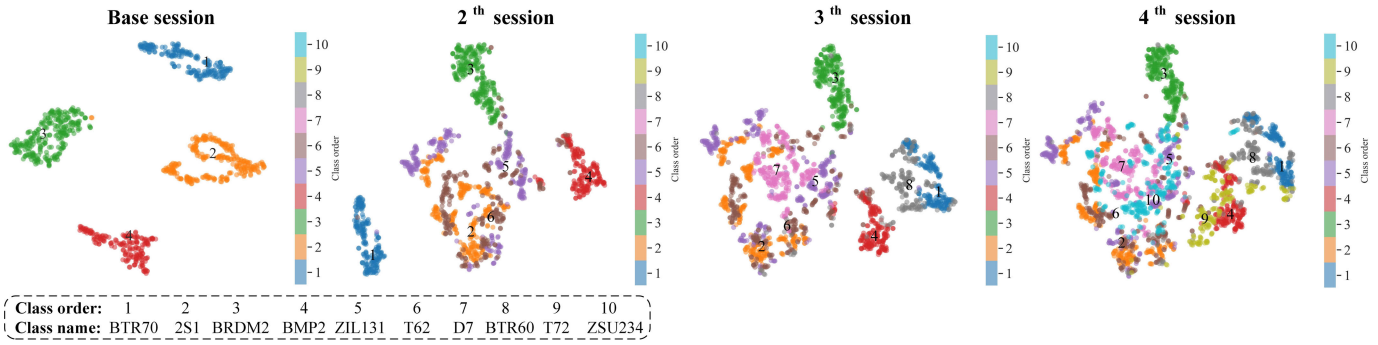


Fig. 21. t-SNE of ten-class semantic features under two-*way* five-*shot* setting.

TABLE VI

COMPARISON WITH STATE OF THE ARTS UNDER ONE-*Way* FIVE-*Shot* SETTING ON THE MSTAR DATASET

Methods	Sessions (%) 1	2	3	4	Avg. Acc	PD ↓
Ft-CNN	98.94	62.31	47.44	38.16	61.71	60.78
Joint-CNN	98.94	77.77	73.9	67.80	79.60	31.14
cRT [53]	98.94	76.49	70.52	63.41	77.34	35.53
iCaRL [36]	92.12	71.14	59.97	48.31	67.88	43.81
LUCIR [37]	99.79	64.18	50.55	41.54	64.02	58.25
ProtoNet [54]	90.85	67.42	57.35	46.88	65.63	43.96
RelationNet [55]	80.64	51.66	40.41	32.62	51.33	48.01
TOPIC [16]	89.86	73.54	61.91	50.89	69.05	38.97
IDLVQC [40]	97.02	70.9	58.67	48.12	68.68	48.90
F2M [43]	85.30	62.38	51.0	41.49	60.04	43.81
CEC [19]	90.75	72.82	66.72	57.95	72.06	32.80
FACT [57]	98.85	74.08	59.47	47.97	70.09	50.87
Ours (CPL)	**99.89**	**76.70**	**69.47**	**58.87**	**76.23**	**41.02**

TABLE VII

COMPARISON WITH STATE OF THE ARTS UNDER ONE-STEP FIVE-SHOT SETTING ON THE MSTAR DATASET

Methods	Sessions (%) 1	2	Avg. Acc	PD ↓
Ft-CNN	98.94	38.27	68.60	60.67
Joint-CNN	99.57	71.27	85.42	28.30
cRT [53]	99.57	70.02	84.80	29.55
iCaRL [36]	92.12	51.86	71.99	40.26
LUCIR [37]	99.79	38.48	69.14	61.31
ProtoNet [54]	90.82	46.92	68.90	43.90
RelationNet [55]	80.56	32.68	56.62	47.88
TOPIC [16]	92.10	51.31	71.70	40.79
IDLVQC [40]	97.02	48.33	72.68	48.69
F2M [43]	85.30	41.14	63.22	44.16
CEC [19]	90.07	58.48	74.27	31.59
FACT [57]	98.85	56.05	77.45	42.80
Ours (CPL)	**99.89**	**58.42**	**79.16**	**41.47**

method at the last several sessions are relatively stable after adaption to new knowledge. Benefiting from the alignment of relationships between class-aware exemplars and prototypes, intrinsically structural mappings of old knowledge can be preserved well, improving the model's stability of feature spaces of old classes. Maybe because serious squeeze of feature spaces, the semantic structures of old classes are highly changeable at initial sessions (see the first four subfigures of Fig. 20 for examples).

3) Serious overlaps exist among features of specific classes sharing similar and common configurations, which causes much interference to our method for discriminating targets accurately. As shown in the last subfigures of Figs. 20–22, features between specific class pairs, such as T62 and 2S1, BTR60 and BTR70, D7, and ZSU234, are ambiguous, due to their similar functions and configurations. However, for most testing samples with the same labels, features of them extracted by our method can be clearly distinguished by our method.

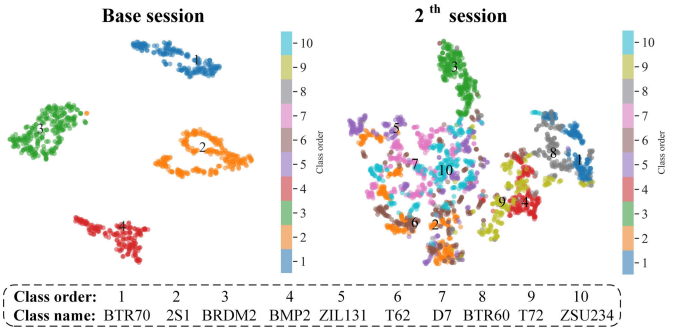


Fig. 22. t-SNE of ten-class semantic features under one-step five-*shot* setting.

G. More Discussion

In this section, more experiments are conducted to evaluate the effects of hyperparameters.

1) Plasticity and Stability Dilemma: During learning, our model's abilities to learn new knowledge and preserve the old are balanced by γ in $\mathcal{L}_{\text{total}}$ of Algorithm 1. The larger the γ, the more the new concepts our model prefers to learn. As shown in Fig. 23, the dilemma reaches a trade-off, and our method achieves the best score on the Avg.Acc when the γ is 1.0. Most specifically, the performance trend can be broadly divided into two stages. At the first stage, where the γ is lower than 1.0, our method's performance in classifying old, new, and total classes is improved persistently with a decrease in the PD rate by releasing its plasticity. It may be because of a moderate balance of the dilemma that representative and general features of new classes can be learned by our method without losing much discrimination on the old. At the second stage, where the γ is over 1.0, the performance decreases or fluctuates. Because of the struggle of our method for learning new concepts from few-shot samples, a serious distortion of learned feature spaces occurs with a weak generalization of the new.

2) Selection of m and N: In the CPL, the ability to learn novel knowledge is controlled by PSS loss, especially the

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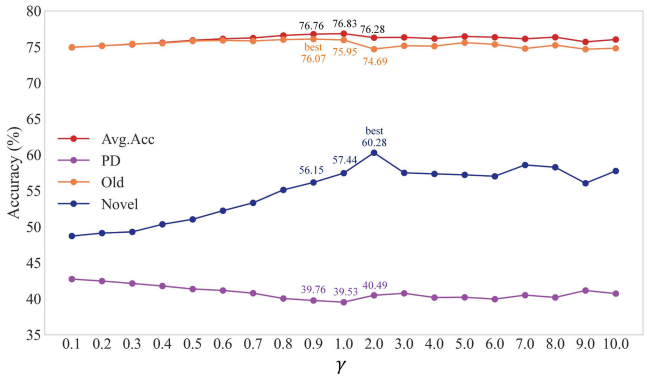


Fig. 23. Performance curves of our method as γ changes. The best results on old and novel classes are achieved when the values of γ are 0.9 and 2.0, respectively. Our method achieves the best performance on the Avg.Acc of all seen categories when the γ is 1.0.

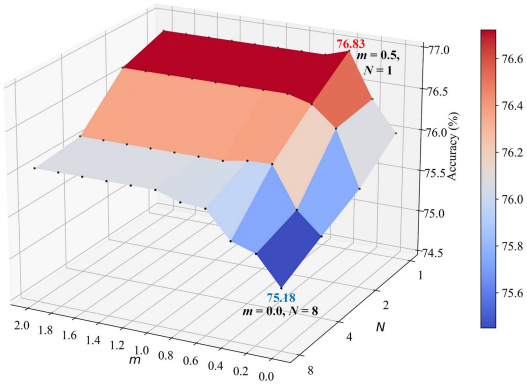


Fig. 24. Avg.Acc of our method at the variation of m and N in PSS loss.

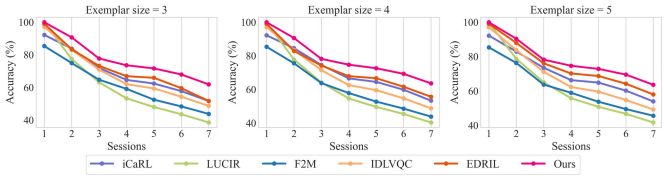


Fig. 25. Avg.Acc of our method compared with exemplar-based methods at the variation of exemplar numbers.

margin m and the number of selected pairs N. To investigate their contributions, the Avg.Acc of our method at different combinations of the two factors is shown in Fig. 24. Obviously, with a fixed value of N, the Avg.Acc comes up sharply, as the m increases from 0.0 to 0.4 and reaches a peak, which is 76.83%. Then, the performance keeps stable and drops slightly, as the m increases continually from 0.4 to 2.0. However, no matter what m our method takes, the performance decreases consistently as N increases. Significantly, the worst performance is achieved (75.18%) when the two factors are 0.0 and 8, respectively. It may be because of biased guidance of few-shot samples of novel classes that the model easily tends to forget the old knowledge during adapting to the new, especially when the two factors reach large values. As a result, the best performance is obtained when the m and the N are set to 0.4 and 1, respectively.

3) Variation of Exemplar: In our method, exemplars are used for knowledge distillation at each incremental session. The larger the number of class-specific exemplars, the richer the supervision is provided for knowledge preservation. The performance (Avg.Acc) of the CPL and the compared exemplar-based benchmarks at various sizes of exemplars is shown in Fig. 25. It can be seen that our method is superior to the compared methods by a large margin at all incremental sessions under all conditions of the number of exemplars. Besides, our method consistently outperforms all these methods, as the number of exemplars increases, verifying the ability of our method to retain the learned knowledge based on limited information.

V. Conclusion

Aiming at the misalignment between the streaming data in real-world SAR ATR scenarios and the closed-world discrimination of data-driven DL-based models, we first unlock FSCIL in the SAR ATR field and propose a CPL framework to address the intrinsic plasticity and stability dilemma and domain-related challenges. In the CPL, target knowledge is represented as class-aware prototypes, and various cosine structural losses are designed inspired by the relationships between azimuth-aware structures of targets and their semantic features under the cosine criterion. Specifically, the model's capability of learning new concepts is provided by the PSS and the PC losses, in which interclass separations and intraclass compactness are improved without losing generalization. The PED loss and PR strategy are designed to preserve the old knowledge, in which relational structures between class-aware exemplars and prototypes are regularized, and the misalignment of feature spaces is smoothed, respectively. At inference, a cosine-based NCM classifier is adopted for evaluation. Extensive experiments conducted on the MSTAR dataset under three settings of FSCIL and elaborated ablation studies demonstrate the competitive performance achieved by our method compared with various benchmarks, verifying its effectiveness for tackling the FSCIL of SAR ATR.

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